

The Dragon Stream Cipher: Design, Analysis, and Implementation Issues

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Abstract. Dragon is a word-based stream cipher. It was submitted to the eSTREAM project in 2005 and has advanced to Phase 3 of the software profile. This paper discusses the Dragon cipher from three perspectives: design, security analysis and implementation. The design of the cipher incorporates a single word-based non-linear feedback shift register and a non-linear filter function with memory. This state is initialized with 128- or 256-bit key-IV pairs. Each clock of the stream cipher produces 64 bits of keystream, using simple operations on 32-bit words. This provides the cipher with a high degree of efficiency in a wide variety of environments, making it highly competitive relative to other symmetric ciphers. The components of Dragon were designed to resist all known attacks. Although the design has been open to public scrutiny for several years, the only published attacks to date are distinguishing attacks which require keystream lengths greatly exceeding the stated 2^{64} bit maximum permitted keystream length for a single key-IV pair.

1 Introduction

The word-based stream cipher Dragon was first presented at ICISC in 2004 [5]. It was also submitted as a candidate to the ECRYPT stream cipher project eSTREAM [12] in April 2005 and has advanced to Phase 3 in the software profile. The objective of eSTREAM is to identify stream ciphers that are suitable for widespread adoption for software applications with high throughput requirements. This paper discusses the Dragon cipher from three perspectives: design, security analysis and implementation.

Stream cipher design has traditionally focussed on bit-based linear feedback shift registers (LFSRs), as these are well studied and produce sequences which satisfy common statistical criteria. Non-linearity is introduced into the keystream of these ciphers either by a non-linear combining function or filter function, or by irregular clocking, or both. When implemented in hardware, bit-based stream ciphers can generate high-throughput keystream, but they are notoriously slow in software. LFSR-based ciphers that use sparse feedback functions may be

vulnerable to attack, but increasing the number of feedback taps decreases the efficiency of the cipher. Finally, the security of some ciphers with linear state update functions is threatened by algebraic attacks [8].

Word-based stream ciphers may exceed the performance of bit-based ciphers in both software and hardware. They produce many times the amount of keystream per iteration than ciphers built using bit-based LFSRs. In software they can outperform even fast block ciphers like the Advanced Encryption Standard [19] by almost an order of magnitude. Although it is easy to assess the speed of these word based stream ciphers, it is difficult to precisely quantify the security they provide.

Dragon was designed with both security and efficiency in mind. It uses a word based non-linear feedback shift register (NLFSR) in conjunction with a non-linear filter to produce keystream as 64-bit words. Dragon has a throughput of gigabits per second and requires little more than four kilobytes of memory, so is suitable for use in many environments including those that are resource-constrained. Not only is Dragon an efficient keystream generator, it is also key agile, making it especially suitable for mobile applications and wireless communications.

Section 2 of this paper presents the specification for Dragon, while Section 3 describes the design decisions behind the algorithm. Sections 4 and 5 provide an analysis of the security related properties of Dragon and discuss the application of current cryptanalytic techniques. Section 6 discusses the performance of Dragon in both software and hardware, and associated implementation issues. Finally, we draw conclusions regarding the security provided by the cipher against all currently known attacks, including both distinguishing and key/state recovery attacks.

2 Dragon Design Specification

Dragon can be used with a 128-bit key and 128-bit IV, or with a 256-bit key and 256-bit IV. We term the former version *Dragon-128*, and the latter *Dragon-256*. Keystream generation for both versions is identical, but key initialization differs slightly.

Both versions of Dragon are constructed using a single 1024-bit word based NLFSR and 64-bit memory M , giving a state size of 1088 bits. The initial state is populated using the key and IV in conjunction with the state's update function F . The update function is also used in keystream generation.

2.1 Dragon's State Update Function (F Function)

Dragon-128 and Dragon-256 both use the same F function. As shown in Figure 1, F is an invertible mapping from 192 to 192 bits: it takes 6×32 bits as input (denoted a, b, c, d, e, f) and produces 6×32 bits as output (denoted a', b', c', d', e', f'). The network consists of three layers: a initial mixing layer, an s-box layer, and a final mixing layer. The mixing layers use addition modulo 2^{32}